

VEDANT PAWAR

Python Developer · AI/ML Engineer · NLP & Backend Developer · LLM Integration Specialist
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PROFESSIONAL SUMMARY

Results-driven Python Developer and AI/ML Engineer with hands-on experience building production-grade NLP systems, LLM-integrated backends, and real-time AI applications. Architected CREVA — a 7-domain autonomous AI robot — using FastAPI, LangChain, RAG pipelines, and Docker, achieving 25% workflow efficiency gains across a 5-engineer team. Proficient in Python, Flask, Django, REST APIs, ChromaDB, OpenRouter, SerpAPI, ElevenLabs, AWS deployment patterns, and CI/CD workflows. Actively seeking Python/AI backend or ML Engineer roles to deliver measurable business impact.

CORE TECHNICAL SKILLS

Languages & Frameworks: Python · FastAPI · Flask · Django · REST API Development · SQL (MySQL · PostgreSQL) · JavaScript · HTML5 · CSS3

AI / ML / NLP: LLM Integration · LangChain · RAG Pipelines · Prompt Engineering · NLP Pipelines · Hugging Face Transformers (basics) · AI Model APIs (OpenRouter, SerpAPI) · Machine Learning · Data Preprocessing · Sentiment Analysis · Text Classification · Vector Databases (ChromaDB)

Voice & Multimodal AI: gTTS · ElevenLabs API · Multilingual Speech Generation · Voice-enabled Interaction

Frontend & UI: Gradio · Streamlit · React (basics)

Libraries & Data: Pandas · NumPy · Matplotlib · JSON · CSV Processing · SQLAlchemy (basics)

DevOps, Cloud & Tooling: Git · GitHub · Docker · Docker Compose · AWS (EC2, S3 — deployment patterns) · CI/CD Pipelines · GitHub Actions (basics) · VS Code · Agile / Scrum

Core CS Concepts: OOP · Data Structures & Algorithms · API Design Patterns · Microservices Concepts · Scalable Architecture · Backend Optimization · Async Programming

PROFESSIONAL EXPERIENCE

AI/ML & IoT Intern — Incubators (CREVA AI Robot Project) Jan 2026 – Mar 2026

Team size: 5 engineers | Stack: Python · FastAPI · NLP · REST APIs · IoT integration

- Architected NLP-based interaction pipelines for the CREVA AI Robot, enabling real-time AI request processing across 7 functional domains and improving overall workflow efficiency by 25% — reducing average task completion time from ~40s to ~30s.
- Integrated 5+ AI modules with web-based interfaces via RESTful APIs, achieving seamless cross-module communication; reduced inter-service API latency by ~30% through payload optimization and connection pooling.
- Designed and implemented a scalable AI + IoT backend architecture supporting modular, hot-swappable deployment workflows, enabling 3 new AI modules to be added without downtime.
- Containerized the FastAPI backend using Docker, streamlining local development and enabling consistent, reproducible deployment environments across the 5-person engineering team.
- Collaborated in an Agile/Scrum environment — delivered across 6 two-week sprints, maintaining 100% on-time sprint delivery while iterating on real-time AI system integration.

KEY PROJECTS

CREVA AI System — Multi-Domain Autonomous AI Robot | Stack: Python · FastAPI · NLP · LangChain · Docker · REST APIs

- Designed and deployed NLP-driven interaction pipelines enabling real-time, context-aware AI communication across 7 independent functional domains — serving 20+ concurrent request types.
- Integrated multiple LLM and AI APIs into a unified, scalable FastAPI backend with LangChain orchestration, supporting hot-swappable modular AI workflows and reducing integration effort by ~40%.
- Engineered modular backend architecture with clean separation of concerns, containerized with Docker — enabling parallel feature development across a 5-engineer team with zero merge conflicts on core APIs.

- Implemented robust API communication with error handling, retry logic, and async request queuing, achieving 99%+ uptime during active testing across 7 functional domains.

AI News Fetch System — Real-Time AI News Aggregation Platform |

github.com/vedantpawar09/AI-News-Project

Stack: Python · FastAPI · SerpAPI · OpenRouter AI · LangChain · ChromaDB (RAG) · Gradio

- Built a real-time news aggregation engine using SerpAPI for content retrieval and OpenRouter LLM models for multi-article summarization — reducing content-to-insight time by ~60% (from ~10 min manual to ~4 min automated).
- Implemented a RAG pipeline using LangChain + ChromaDB vector database for semantic article retrieval and context-aware summarization — improving summary relevance scores by ~35% vs. naive LLM prompting.
- Designed a high-throughput async FastAPI backend with concurrent request handling, processing up to 15 simultaneous news feed queries without performance degradation.
- Containerized the full stack with Docker, enabling one-command local deployment; stored article embeddings in a PostgreSQL-backed metadata layer for persistent retrieval.

Advanced NLP System — Voice + Multilingual AI Platform | github.com/vedantpawar09/NLP_Project

Stack: Python · Flask · FastAPI · gTTS · ElevenLabs API · PostgreSQL · Docker

- Engineered a production-grade multilingual NLP system supporting voice-enabled interaction with real-time speech synthesis across 5+ languages, serving 100+ test users in validation rounds.
- Integrated gTTS and ElevenLabs APIs for natural-language speech generation; stored user session data and language preferences in a PostgreSQL database with SQLAlchemy ORM.
- Built a scalable API-first backend using Flask and FastAPI with async processing, handling multilingual tokenization, language detection, and response generation pipelines at <200ms median latency.
- Containerized with Docker Compose (app + PostgreSQL services), reducing onboarding time for new contributors from 2 hours to under 15 minutes.

EDUCATION

Bachelor of Computer Applications (BCA)

2022 – 2025

Shri Shivaji Science College, Amravati, Maharashtra

CERTIFICATIONS & CONTINUOUS LEARNING

- Python for Everybody — Coursera
- FastAPI & REST API Development — Self-directed (3 production projects)
- LangChain for LLM Application Development — DeepLearning.AI (In Progress)
- Machine Learning & NLP Fundamentals — Applied via CREVA AI and NLP System projects
- AWS Cloud Practitioner Essentials — In Progress (EC2, S3, Lambda deployment patterns applied in projects)
- Docker & Containerization — Applied across all 3 production GitHub repositories
- Git & GitHub Version Control — Active usage across 3+ production-grade repositories

ADDITIONAL INFORMATION

- Strong focus on AI Engineering, LLM integration, RAG systems, scalable backend architecture, and production-ready AI automation.
- Hands-on experience with LangChain, ChromaDB, vector databases, Docker, and async FastAPI — skills directly aligned with 2026 AI/ML job market demand.
- Open to remote, hybrid, and startup environments across Pune, Mumbai, Nashik, and pan-India — fast learner with demonstrated self-driven delivery record across 3 independent AI projects.
- GitHub: All 3 projects are public with documented READMEs — available for immediate portfolio review at github.com/vedantpawar09